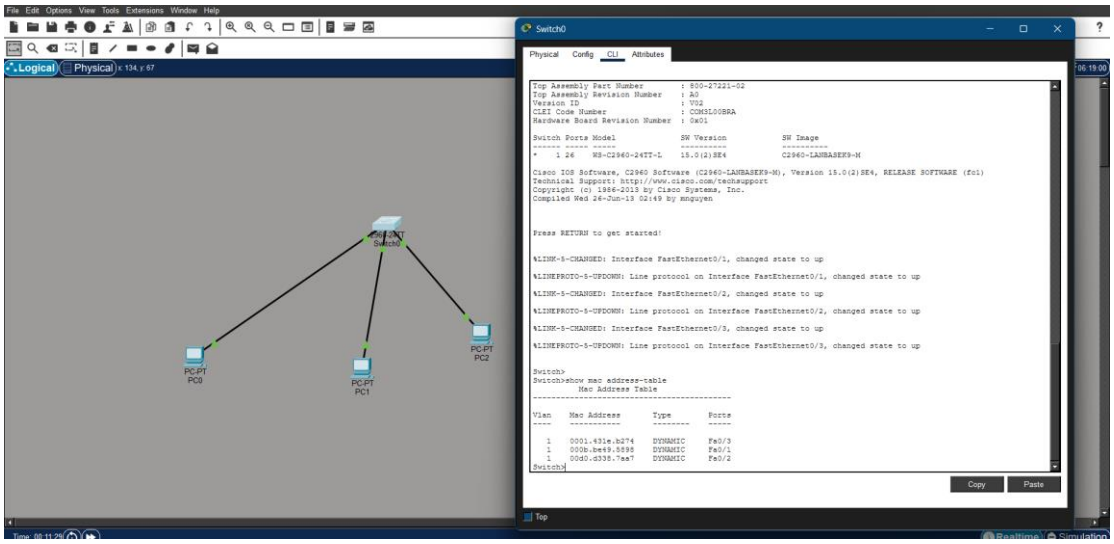
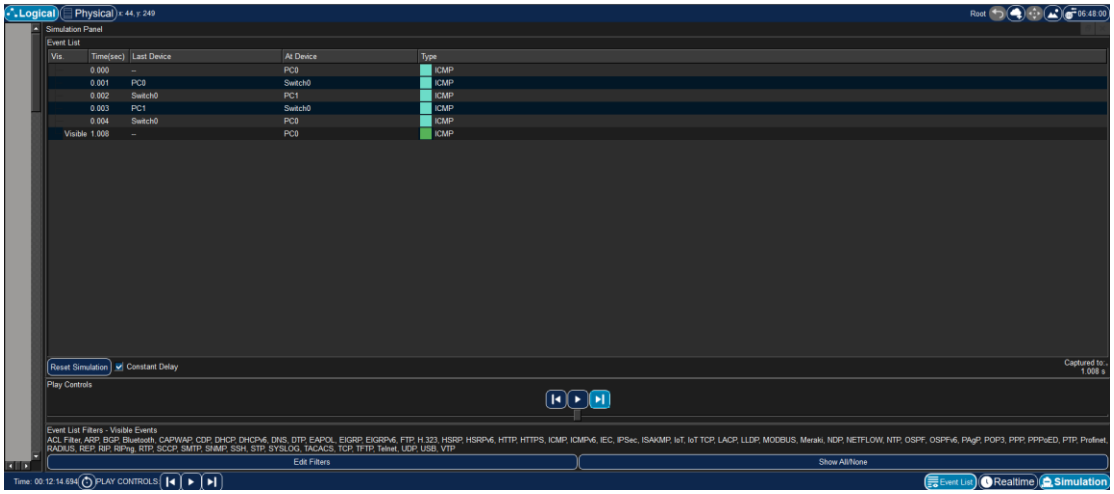
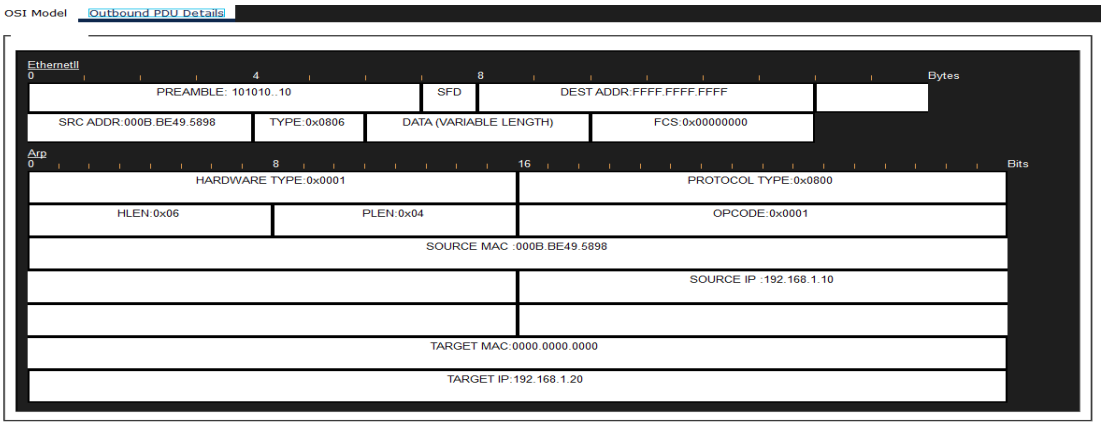
1)



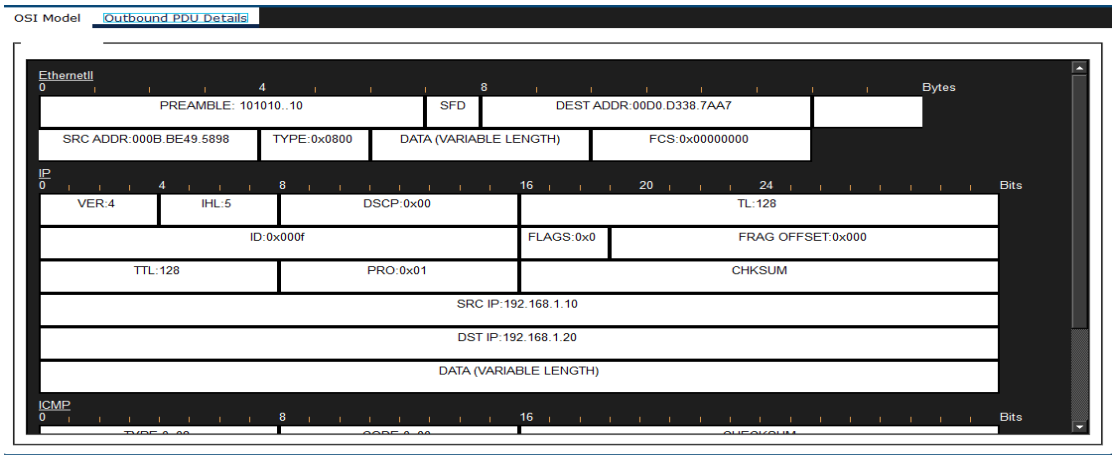


2)



To complete this task, I used Cisco Packet Tracer's Simulation Mode to capture and analyse Ethernet frames between PC0 (192.168.1.10) and PC1 (192.168.1.20) connected through a 2960 switch. The

first frame captured was an ARP Request. The Destination MAC was FFFF.FFFF.FFFF (broadcast), Source MAC was 000B.BE49.5898 (PC0), and the Ethertype was 0x0806 identifying it as an ARP packet. The ARP payload showed Opcode 0x0001 (Request), Source IP 192.168.1.10, Target IP 192.168.1.20, and Target MAC 0000.0000.0000 because PC0 did not yet know PC1's MAC address. Since it was a broadcast, the switch flooded this frame to all ports and only PC1 responded.



- After PC0 learned PC1's MAC address from the ARP reply, it sent an ICMP Echo Request frame. This time the Destination MAC was 00D0.D338.7AA7 (PC1's actual MAC), Source MAC was 000B.BE49.5898 (PC0), and the Ethertype was 0x0800 identifying it as an IPv4 packet. The IP payload showed Source IP 192.168.1.10, Destination IP 192.168.1.20, TTL 128, and Protocol 0x01 confirming ICMP. Since the destination MAC was now a specific unicast address, the switch forwarded the frame directly to PC1's port only without any flooding. This task demonstrated how ARP resolves MAC addresses before communication begins and how the Ethertype, MAC addresses, and FCS fields each play a specific role in Ethernet frame delivery.

3)

```
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1492
        inet 172.17.83.38  netmask 255.255.240.0  broadcast 172.17.95.255
        inet6 fe80::215:5dff:feb6:15d7  prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:b6:15:d7  txqueuelen 1000  (Ethernet)
        RX packets 22638  bytes 23468226 (23.4 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 12190  bytes 861345 (861.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 377  bytes 45945 (45.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 377  bytes 45945 (45.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip addr add 192.168.1.10/24 dev eth0
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:b6:15:d7 brd ff:ff:ff:ff:ff:ff
    inet 172.17.83.38/20 brd 172.17.95.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.1.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:feb6:15d7/64 scope link
       valid_lft forever preferred_lft forever
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip link eth0 down
Command "eth0" is unknown, try "ip link help".
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip link set eth0 down
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip link set eth0 down address 02:AA:BB:CC:DD:10
```

```
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip link set eth0 up
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip link sho eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ping -c 8.8.8.8
ping: invalid argument: '8.8.8.8'
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ping -c 4 8.8.8.8
ping: connect: Network is unreachable
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=3.87 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.038 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.035 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3043ms
rtt min/avg/max/mdev = 0.035/0.998/3.872/1.659 ms
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1492
        inet 172.17.83.38  netmask 255.255.240.0  broadcast 172.17.95.255
        inet6 fe80::aa:bbff:fecc:dd10  prefixlen 64  scopeid 0x20<link>
        ether 02:aa:bb:cc:dd:10  txqueuelen 1000  (Ethernet)
        RX packets 22664  bytes 23471742 (23.4 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 12214  bytes 863105 (863.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

4)

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert 192.168.1.10

Tracing route to 192.168.1.10 over a maximum of 30 hops:

  1    *        *        *        Request timed out.
  2    *        *        *        Request timed out.
  3    *        *
Control-C
^C
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

5)

Linux Kernel Ethernet Support

The Linux kernel supports Ethernet communication through a layered networking subsystem. According to kernel.org, every network interface is represented by a data structure called struct net_device defined in include/linux/netdevice.h, which stores the interface name, MAC address, MTU size, and operational state flags like UP and RUNNING. When an Ethernet frame arrives, the kernel allocates a socket buffer called sk_buff, passes it up the stack using netif_receive_skb(), and inspects the Ethertype field to route it to the correct protocol handler such as ip_rcv() for IPv4 or arp_rcv() for ARP. For outgoing frames, the kernel builds the sk_buff, attaches the correct MAC addresses, and calls ndo_start_xmit() to pass it to the hardware. NAPI improves performance at high speeds by switching from interrupt mode to batch polling, reducing CPU overhead significantly. The kernel also exposes NIC settings and statistics through the ethtool interface for monitoring and configuration.

6)

```
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1492
        inet 172.17.83.38  netmask 255.255.240.0  broadcast 172.17.95.255
        inet6 fe80::aa:bbff:fecc:dd10  prefixlen 64  scopeid 0x20<link>
        ether 02:aa:bb:cc:dd:10  txqueuelen 1000  (Ethernet)
        RX packets 22664  bytes 23471742 (23.4 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 12214  bytes 863105 (863.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
    inet 172.17.83.38/20 brd 172.17.95.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.1.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::aa:bbff:fecc:dd10/64 scope link
       valid_lft forever preferred_lft forever
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip addr add 192.168.56.10/24 dev eth0
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
    inet 172.17.83.38/20 brd 172.17.95.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.1.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.56.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::aa:bbff:fecc:dd10/64 scope link
       valid_lft forever preferred_lft forever
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip route show
172.17.80.0/20 dev eth0 proto kernel scope link src 172.17.83.38
```

```
root@EURLTP-489: /mnt/c/Us   Windows PowerShell

root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
    inet 172.17.83.38/20 brd 172.17.95.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.1.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::aa:bbff:fecc:dd10/64 scope link
       valid_lft forever preferred_lft forever
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip addr add 192.168.56.10/24 dev eth0
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
    inet 172.17.83.38/20 brd 172.17.95.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.1.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.56.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::aa:bbff:fecc:dd10/64 scope link
       valid_lft forever preferred_lft forever
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip route show
172.17.80.0/20 dev eth0 proto kernel scope link src 172.17.83.38
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.10
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.10
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip link set eth0
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip link set eth0 down
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# sudo ip link set eth0 up
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip link show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
```

Summary:

To configure a basic LAN interface in Linux, I used the ip command which is the modern replacement for ifconfig as documented at kernel.org. I ran ip addr show eth0 to view the current interface state which showed the MAC address, existing IP address, and that the interface was UP. I then assigned a new static IP address using sudo ip addr add 192.168.56.10/24 dev eth0 and verified it was added successfully by running ip addr show eth0 again which showed three active IP addresses on the same interface. Finally I confirmed the routing table using ip route show which automatically showed new network routes for each assigned IP, and restarted the interface using sudo ip link set eth0 down and sudo ip link set eth0 up to confirm full connectivity was restored.

7)

ip link show

ip link add name br0 type bridge

ip link set br0 up

ip link set eth0 master br0

ip link show master br0

```
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip link add name br0 type bridge
ip link set br0 up
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
3: br0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default qlen 1000
    link/ether 96:47:d3:d1:0b:9e brd ff:ff:ff:ff:ff:ff
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip link set eth0 master br0
root@EURLTP-489:/mnt/c/Users/varssh/Documents/linux_assgn# ip link show master br0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1492 qdisc mq master br0 state UP mode DEFAULT group default qlen 1000
    link/ether 02:aa:bb:cc:dd:10 brd ff:ff:ff:ff:ff:ff permaddr 00:15:5d:b6:15:d7
```

Conclusion:

Ethernet communication in a LAN is achieved through ARP-based address resolution and MACbased frame forwarding, while the Linux kernel efficiently manages packet processing through its networking stack.